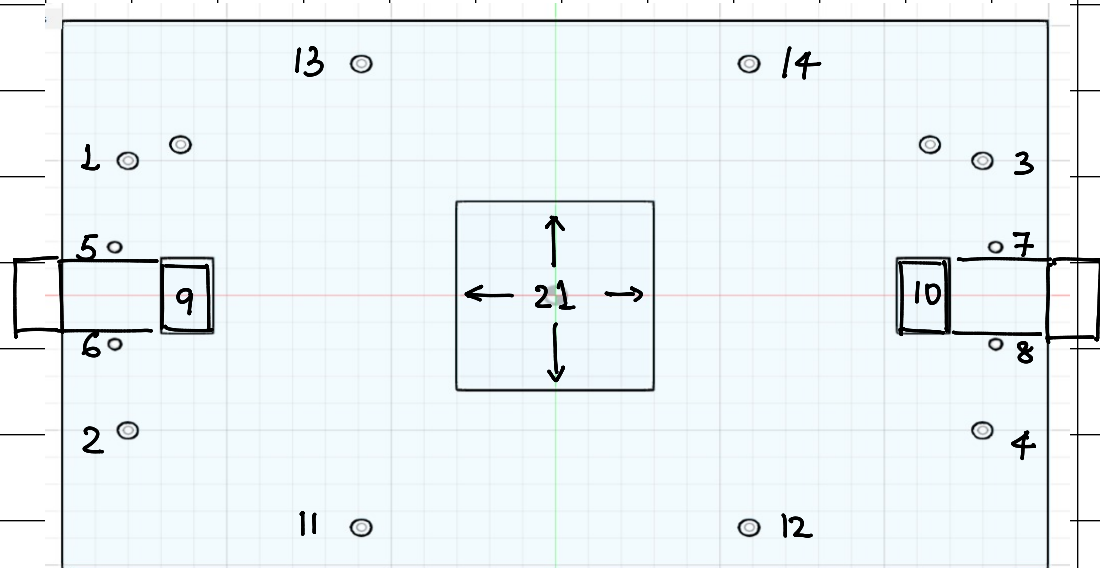
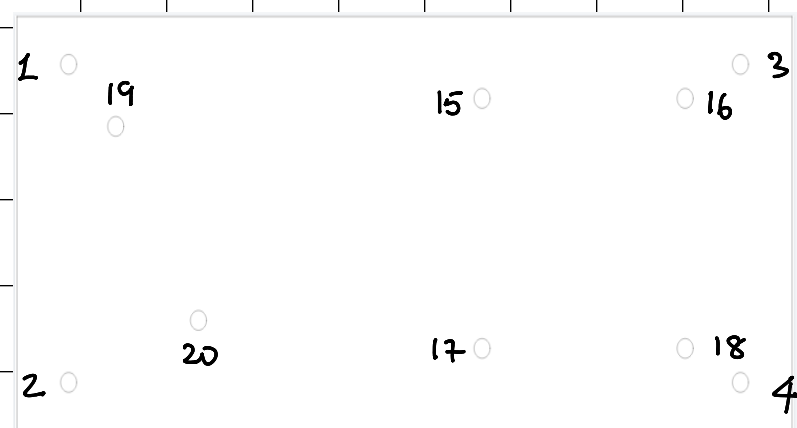


Assembly Instructions:-

① Mechanical assembly:-



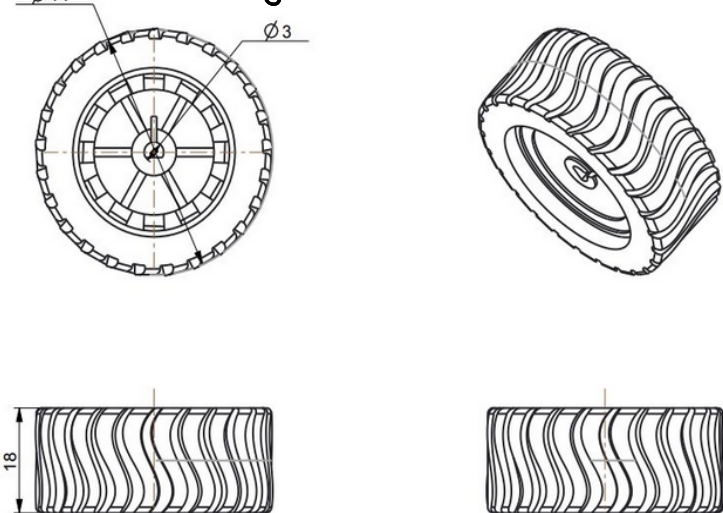
- make cuts into your acrylic sheet in accordance to dxf file (Level0.dxf)
- (1,2,3,4) are holes where you use hex nut to screw 4 hex spacers and 4 more on the 4 screwed hex spacers.
- (5,6,7,8) are holes for motor mounting bracket (9,10) are slots for connector interface to motor.
- (11,12,13,14) are holes for screwing in the line follower sensor module.



- mount the motor driver, buck converter and voltage meter on the second sheet according to the dxf file (Level1.dxf)

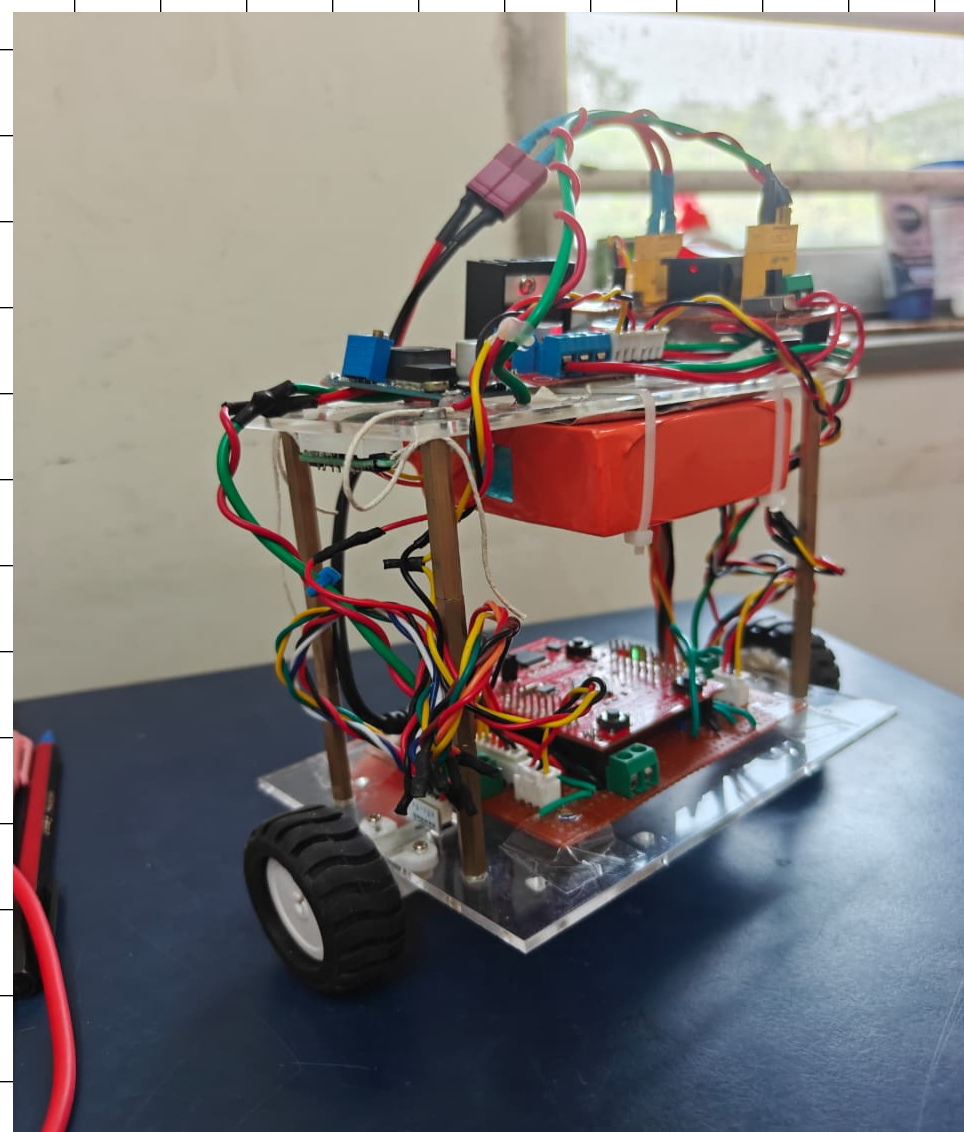
- 3D print 2 wheels following the schematic

→ something greater for smaller vibrations



Components Required:-

- 2x N20 DC motor with Encoder (12V, 120 RPM)
- 2x N20 motor mounting bracket
- 3mm Acrylic sheet
- 8x M3 Brass spacers 35 mm
- 1x MPU9250
- 1x RLS08 line follower sensor
- 2x JST-XH 2.54 mm (01x06) male
- 2x JST-XH 2.54 mm (01x03) male
- 4x JST-XH 2.54 mm (01x06) female
- 4x JST-XH 2.54 mm (01x03) female
- 1x JST-XH 2.54 mm (01x10) male
- 1x JST-XH 2.54 mm (01x10) female
- 1x L298N motor driver module to Jumper female
- 1x LM2596S DC-DC Buck converter module
- 1x 0.28 in 0-100V 3wire voltage display
- 1x 11.1V 3S Li-ion Battery 2200mAh
- 2x Slide switch
- 1x XT-60 female to Nylon-T male
- 2x XT-60 male connector
- 1x XT-60 Female cable
- 1x micro-USB cable
- 1x SPST Rocker switch



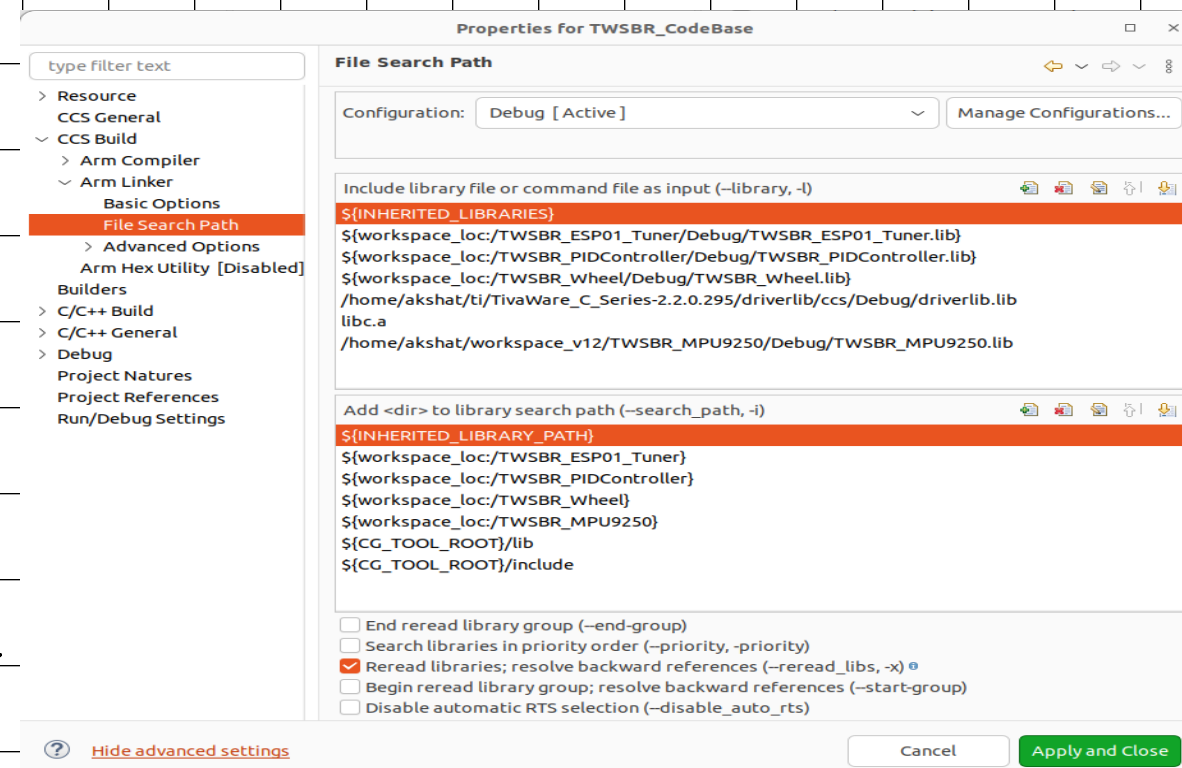
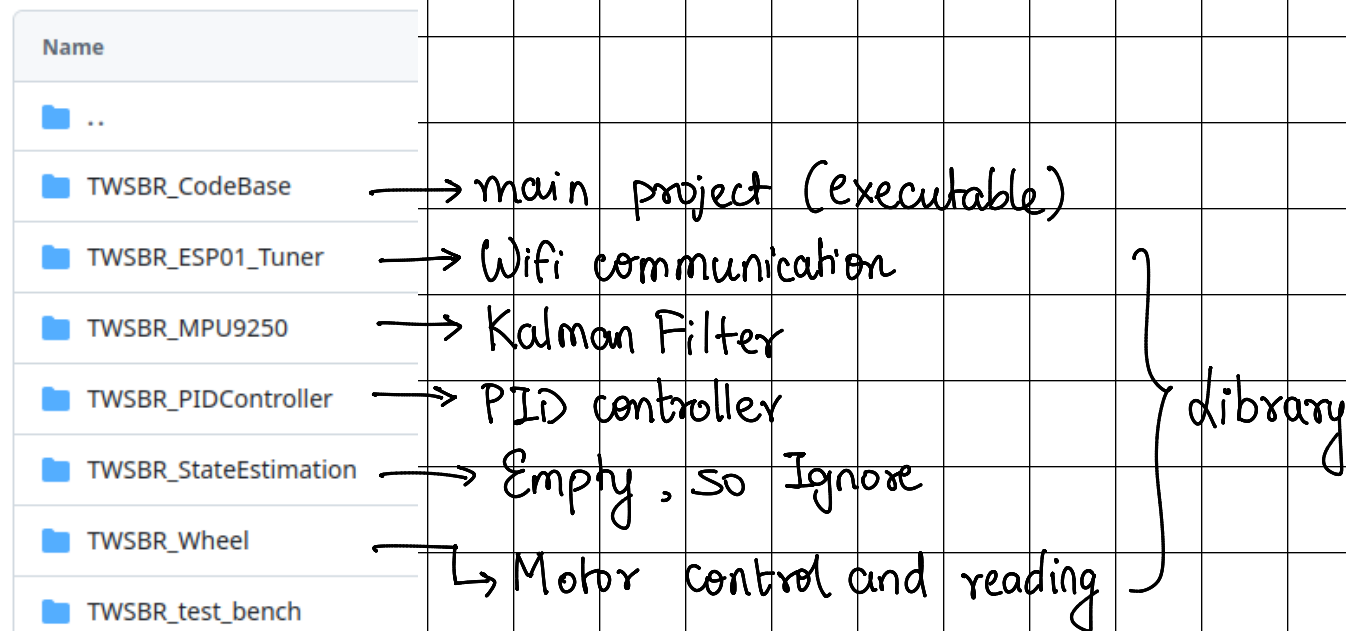
② Electrical Assembly:- Use schematic file in Electrical Subsystem.

Follow the schematic file to make a PCB such that 10 pin JST-XH male and 10 pin berg connector (female) fit in cavity labeled "21".

③ Software Subsystem:-

① Download repository and import projects in "TWSBR_Code_Base" folder to CCS workspace.

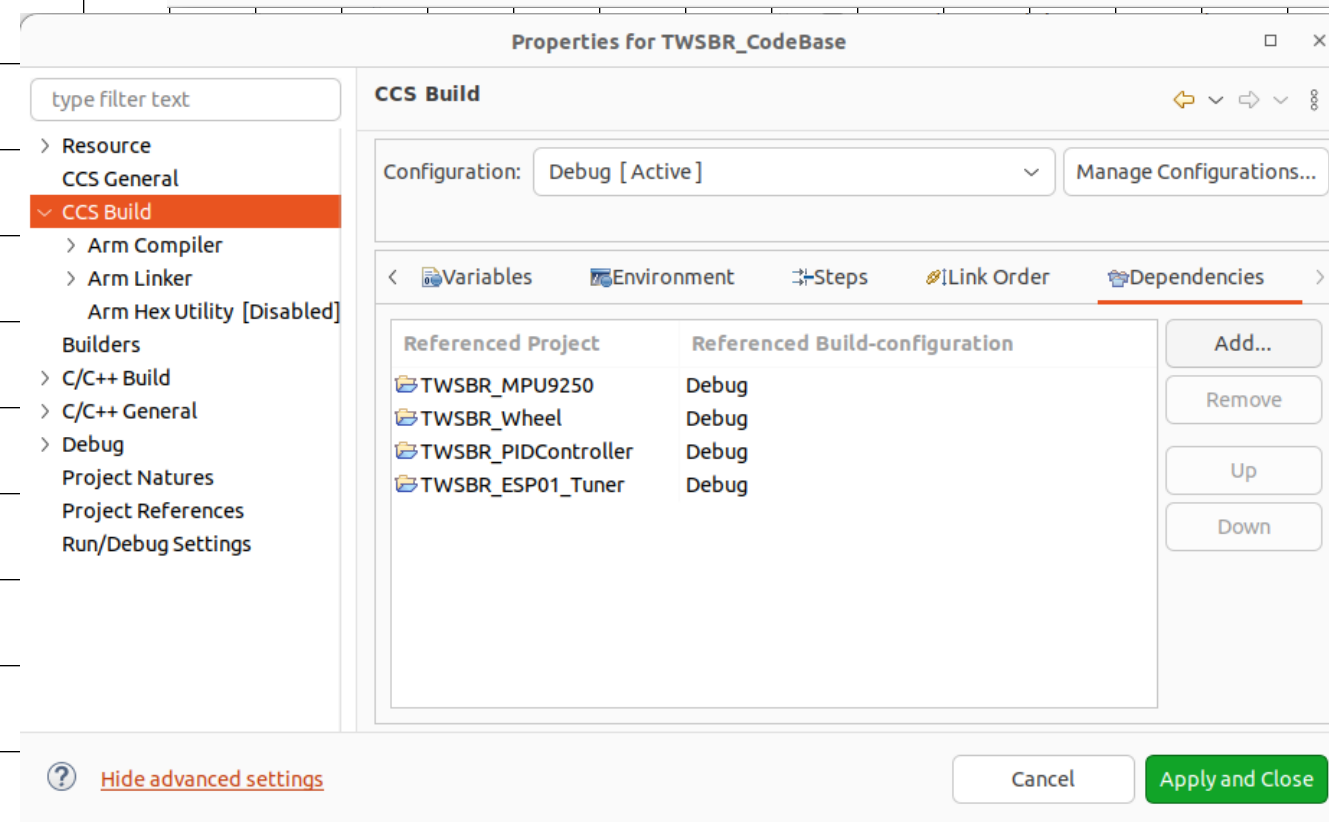
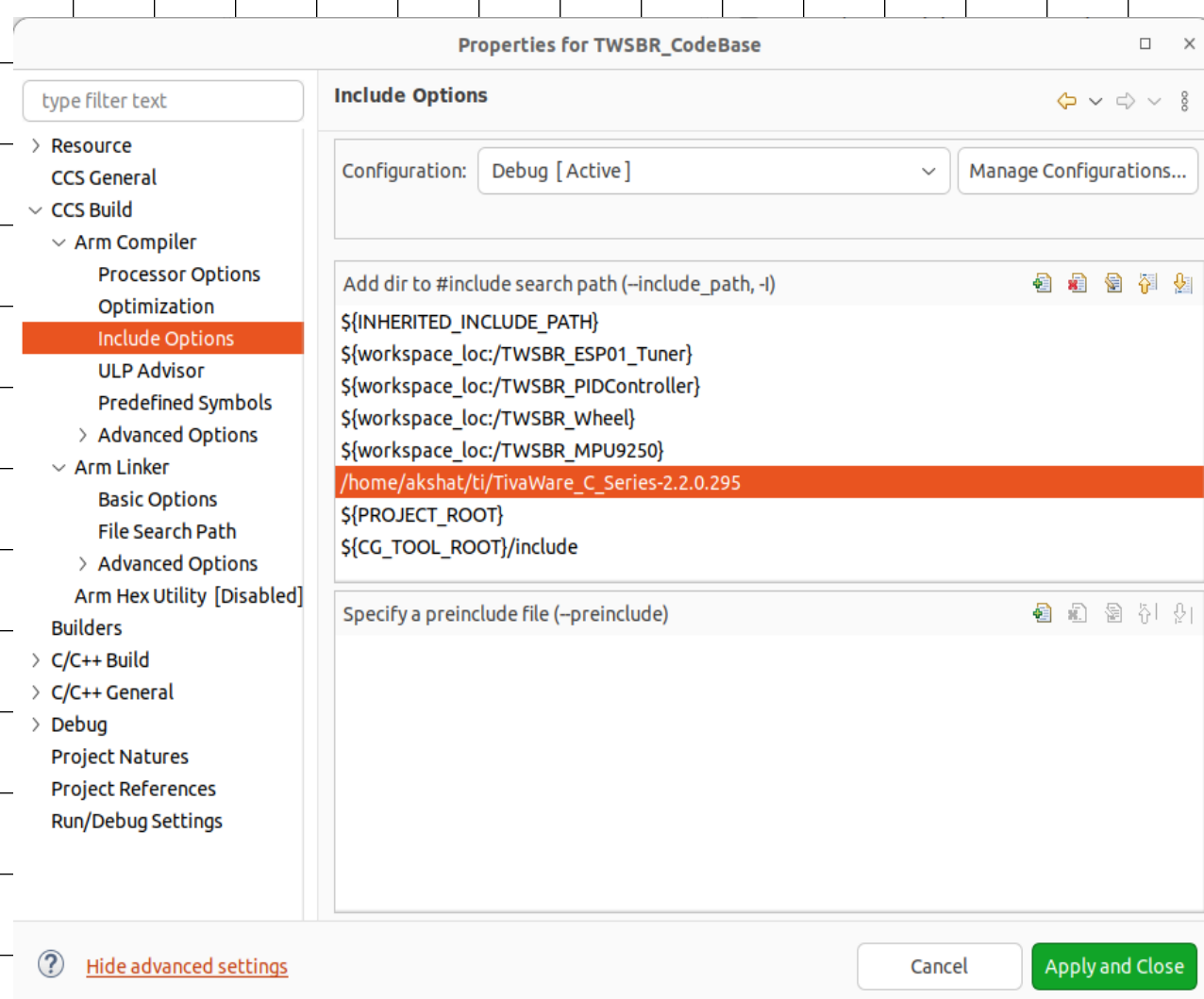
②



a) Change include path for TivaWare

b) Add library file (.lib) to -l of executable project.

c) Add library projects to executable project as dependencies



This should enable you to build the project